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Michael Ruloff & Dominik Petko

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School principals' educational goals and leadership styles for digital transformation: results from case studies in upper secondary schools

Michael Ruloff ^a and Dominik Petko ^b

^aSchool of Education, University of Applied Sciences and Arts Northwestern Switzerland, Windisch, Switzerland; ^bInstitute of Education, University of Zurich, Zurich, Switzerland

ABSTRACT

Digital technologies are transforming the job market and pose new challenges to education, yet there is little research that specifically examines how school principals' goals and leadership styles influence digital development in upper secondary schools. The present study explores how their ambitions and educational goals affect their approaches to leadership for digital transformation in their schools. The study employs an exploratory and qualitative research approach based on interviews with a full sample of nine upper secondary school principals from Central Switzerland. Transcripts have been analyzed using qualitative content analysis. The findings suggest that all principals express a clear orientation toward student-oriented teaching when integrating technology in their schools. However, there are substantial differences in terms of leadership processes to address this change: One principal oriented toward transformational leadership reported more rapid and more profound changes than all the principals oriented toward a more transactional type of leadership. In this case, transformational leadership is associated with faster implementation of digital technologies and also with a clearer focus on educational goals rather than on tools. This study expands the knowledge on principals' technology leadership practices as it provides insights into school leaders' attitudes and concepts and highlights the benefits of transformational leadership.

Introduction

Digital transformation is one of the most significant challenges for education in the 21st century (Voogt et al., 2013, 2018). This goes well beyond the integration of technologies, requiring changes with regard to school curricula and the overall culture of teaching and learning, in order to prepare the next generation for different job markets and societies. Switzerland is a special case in this regard. Although it often gains top marks in world-wide competitiveness rankings with regard to digital transformation, related school development lags behind. Studies on technology-rated school development in the Swiss educational context show that, on average, digital technologies are not used in class on

CONTACT Michael Ruloff  michael.ruloff@fhnw.ch  School of Education, University of Applied Sciences and Arts Northwestern Switzerland, Windisch 5210, Switzerland

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a regular basis compared to other countries (OECD, 2015). Additionally, compared internationally, Swiss teenagers in the 8th grade show average digital competencies (Eickelmann & Drossel, 2016). The Swiss case is remarkable: As a rich and technologically advanced industrial nation, Switzerland is one of the few countries where the use of computers at school has a negative effect on students' digital competencies (Eickelmann, Schaumburg, Drossel, & Lorenz, 2014). Past studies from Switzerland have indicated that technology integration depends not only on the provision of a sufficient technological infrastructure but also on teachers' technology-related skills and beliefs (Petko, 2008, 2012). While similar findings have also been shown in numerous international studies (Ertmer & Ottenbreit-Leftwich, 2010; Knezek & Christensen, 2016; Niederhauser & Lindstrom, 2018), it seems that teachers in Switzerland are especially reluctant to acknowledge the potential of digital technologies to support student learning (Prasse et al., 2017).

Based on these findings, research has shifted its attention toward the factors influencing teachers' technology-related skills and beliefs. With regard to professional development opportunities, teachers in Switzerland only receive limited support compared to other countries (Fraillon et al., 2014). Instead, research has shown that in Switzerland teacher readiness to use technology is strongly dependent on school readiness (Petko et al., 2018). Core aspects of school readiness include infrastructure (the perceived quality of the hardware and software available as well as technology integration in general), professional development opportunities (teachers' formal and informal exchange as well as more formal courses and training) and leadership (goal clarity and school principal support). As school principals are tasked to take on a leading role in all matters of school development and are able to influence and shape most contextual factors, they have been considered key players for technology integration in schools (Dexter & Richardson, 2020; Dexter et al., 2016; Tan, 2010). This claim is supported by general research into the impact of school principals' leadership behavior, which has substantial impact on teacher well-being and teacher practices, which in turn affect student performance (Daniëls et al., 2019; Liebowitz & Porter, 2019).

Goal orientation of technology-related school leadership

In general, leadership can be understood to be the process of influencing a group of people to achieve a shared goal (Northouse, 2016; Yukl, 2012). Digital leadership goes well beyond the provision of hardware and software to include an alignment of goals with organizational structures, technological tools and capacity building within the workforce so that the entire change is perceived as seamless and meaningful (Cortellazzo et al., 2019). Similar recommendations have been made regarding technology-related leadership in schools. School leaders are expected to clarify the underlying goals prior to other steps of implementation like adapting the curriculum, convincing and empowering teachers to adjust their teaching accordingly, and providing suitable support, professional development opportunities and infrastructure so that technology becomes a meaningful part of the school culture (Dexter, 2018; Dexter et al., 2016; McLeod, 2018; Tan, 2010). In the United States, there are standards regarding successful school leadership (e.g. the ISTE standards for educational leaders or school administrators). Here, too, having a clear goal or a vision

features top of the list. The numerous technology-related leadership duties of school principals have been grouped into three broad types of activities: School principals 1) set directions (develop a joint vision, create a shared meaning, develop expectations, communicate goals), 2) foster professional development (individually and collectively, also by leading by example) and 3) develop the organization (by allocating resources, adapting structures and regulations, creating a joint culture of innovation and reaching out to external stakeholders; Dexter, 2018). More recently, this has been supplemented with two additional aspects: facilitating high-quality learning experiences for students and connecting with external partners (Dexter & Richardson, 2020). Although principals' leadership might be supplemented with distributed forms of leadership by technology coordinators, teacher leaders or even student leaders (Christensen et al., 2018; Riel & Becker, 2008; Wenner & Campbell, 2017), principals still bear the primary responsibility for the process of digital development in their school. Case studies from pioneering schools have repeatedly shown that technology integration works best when technology is implemented as a means to a pedagogical rather than a technological end (Kozma, 2003; Law et al., 2008; Venezky & Davis, 2002). Although there is widespread agreement that goal clarity (ideally a shared vision that was developed together with all stakeholders) is key to the successful digital transformation, it is less clear what types of goals are promising under different circumstances and what kinds of leadership activities are needed to achieve more or less ambitious goals. Setting realistic rather than idealistic goals while considering the restrictions of the local context can be a challenge for school principals (Kotok & Kryst, 2017).

Transformational practices in technology-related school leadership

Despite the unanimous emphasis of the importance of school principals' leadership with regard to technology integration in education, there are only rough guidelines on how to implement and maintain the change process. The literature around school leadership commonly distinguishes between rather top-down approaches, like instructional and transactional leadership, and rather bottom-up approaches, like transformational and distributed leadership (Daniëls et al., 2019; Leithwood, 1992; Leithwood & Jantzi, 2000; Sun & Leithwood, 2012). Research shows that any kind of active school leadership is better than 'laissez-faire' leadership or 'muddling through'. However, the question whether top-down or bottom-up leadership is more successful has remained a constant challenge. Although research on the impact of leadership on student outcomes seems to favor instructional leadership (Robinson et al., 2008; Shatzer et al., 2014), more substantial changes in teaching practices seem to favor combined approaches that include transformational, collaborative or distributed leadership practices (Day et al., 2016; Heck & Hallinger, 2009; Leithwood & Jantzi, 2006; Leithwood et al., 2020; Pietsch & Tulowitzki, 2017; Sun & Leithwood, 2012). Transformational leadership is particularly interesting in this regard. It can be described as a combination of strong bottom-up processes with supportive top-down activities tailored to the needs of the situation at hand. The concept of transformational (and transactional) leadership was first described by Burns (1978) in the context of his research on political leadership. While transformational

leadership is associated with strategies that foster intrinsically motivated change, transactional leadership is focused on incentives and extrinsically motivated change. Bass (1990) described that transactional leadership contains rewards as well as management by expectation (active and passive). Transformational leaders are portrayed as charismatic models of change (providing a vision and a sense of a mission), inspirational, able to stimulate intellectually (encouraging creative thinking, involving stakeholders in active decision-making) and able to consider individual needs (creating a sense of belonging, supporting employees). Meta-analyses of studies from non-educational contexts have shown that transformational leadership leads to greater and more sustainable change in companies and organizations than transactional leadership (Dumdum et al., 2013; Judge & Piccolo, 2004). However, there have been relatively few empirical studies investigating the specifics of transformational leadership for technology integration in schools. While many case studies were able to show positive effects of transformational and participatory leadership on technology integration (Afshari et al., 2009; Hadjithoma-Garstka, 2011; Hauge & Norenes, 2015; Lindqvist, 2019; Ng, 2008; Schrum & Levin, 2013; Yuen et al., 2003), survey-based studies were more ambiguous and highlighted the importance of a combination of bottom-up and top-down leadership activities (Petko et al., 2015; Dexter & Richardson, 2020; Gençer & Samur, 2016). Instead of treating transformational and transactional approaches as opposites, both approaches seem to offer building blocks for a continuum of leadership styles (Sun & Leithwood, 2012). Bass (1985) argued that outstanding leaders are both transformational and transactional. Nevertheless, this raises the question of the alignment of these gradually different leadership styles to different goals and contexts.

Deep change as a result of technology-related school leadership

In recent years, the goals of educational technology integration have shifted dramatically. After decades where technology integration was focused on the provision of hardware and software, the main focus has substantially shifted toward teaching and learning with and about technology (Eickelmann, 2018; Ottestad & Gudmundsdottir, 2018). Furthermore, technology integration in education is expected to address the challenges of the broader digital transformation in society and to foster both digital literacy and 21st century skills (Voogt et al., 2013). As a consequence, current models of technology integration tend to regard more fundamental changes as more appropriate. Models like the Substitution-Augmentation-Modification-Redefinition Model (SAMR; Puentedura, 2012, 2015) or the Replacement-Amplification-Transformation-Model (RAT; Hughes et al., 2006) imply that innovation is more substantial if technology is not only a substitute or a slight improvement for old practices but also empowers educators and students to do new things. With regard to SAMR, it has been criticized more recently that innovation cannot be considered a value in its own right and that the transitivity of the model has not been proven empirically (Hamilton et al., 2016). Instead of considering these stages to be opposites, digital transformation may in fact happen in meaningful ways on all stages. Also, it is unclear how the level of educational change relates to goal orientation and educational leadership practices.

Research question

Principals' leadership is crucial for digital transformation in schools. Research has shown that school development goals, leadership practices and the level of technology integration achieved might be related, but the exact interplay of these factors needs to be explored in further detail. This study therefore aims to analyze principals' views on that complex interplay of these factors. It is guided by the following two key questions: How are upper secondary school principals' technology integration goals related to their leadership practices? And how do their leadership practices affect the level of technology integration in relation to teaching and learning in their schools? In particular, we were expecting to observe a positive interrelation between high-level goals, transformational leadership and deep changes. On the other hand, deviations from this blueprint will nevertheless be given special consideration.

Methods

The study presented in this article was conducted in central Switzerland. It is based on the full sample of all nine upper secondary schools of this specific region, a region with a high density of upper secondary schools.

Sample and context

The sample consists of six public schools and three private schools. Four of the six public schools are vocational baccalaureate schools. The other two schools are general baccalaureate schools as are the three private schools. General baccalaureate schools may also include specialized baccalaureate programs. All of these schools are comparable to high schools. In the Swiss education system, all of the mentioned schools qualify for admission to university (International Standard Classification of Education [ISCED], Level 3a). Students with a vocational baccalaureate are allowed to enlist to university after having passed a test. Public schools are very well funded in Switzerland and they offer high quality education. The private secondary schools involved in the study have a long tradition: They were historically funded and run by the catholic church. As all schools have to adhere to the same cantonal and national regulations and respective curricula, there is a high level of comparability between these schools. During the period of our study, schools were preparing for the integration of computer science and information literacy into the curriculum.

Data collection

The study employs an exploratory research design. To get an insight into school leadership practices, all nine principals of the relevant schools were interviewed individually in their offices at their schools. The interviews were conducted in pairs (one interviewer and one principal). They were conducted over a time span of five weeks. Every interview lasted 30 to 45 minutes and was recorded and transcribed verbatim. Each principal was informed and agreed to the interview being recorded prior to the meeting. They also agreed that their statements might be used in a research study about principals'

leadership in upper secondary schools. The interviews were problem-centered (Witzel, 2000). Problem-centered interviews take the individual situation of each school and principal into account: They enable focusing not only on principals' attitudes toward digital technology but also on their thinking, ideas, guiding principles, goals and visions. Thus, principals had the opportunity to freely express their perspective and their views (Witzel, 1985; Witzel & Reiter, 2012). The basic structure of the interviews included standardized questions: Principals had to characterize their school and describe the status and use of digital technology in their institution. The main part of the interview related to personal digital experience and leadership practices regarding technology integration in education. Principals were asked specifically about their ambitions and goals, as well as their leadership and the procedures in relation to digital development in their schools. Furthermore, they were asked about use of digital technology at their school as well as digital integration in general.

Analysis

The interviews were professionally transcribed and evaluated using qualitative content analysis (Mayring, 2015). The transcribed interviews were analyzed individually (Within-case analysis) as well as compared with one another (Cross-case analysis). Each transcript was imported into MAXQDA and coded along a coding scheme (described below) developed on the basis of the research questions and the theoretical considerations presented in the introduction (Figure 1).

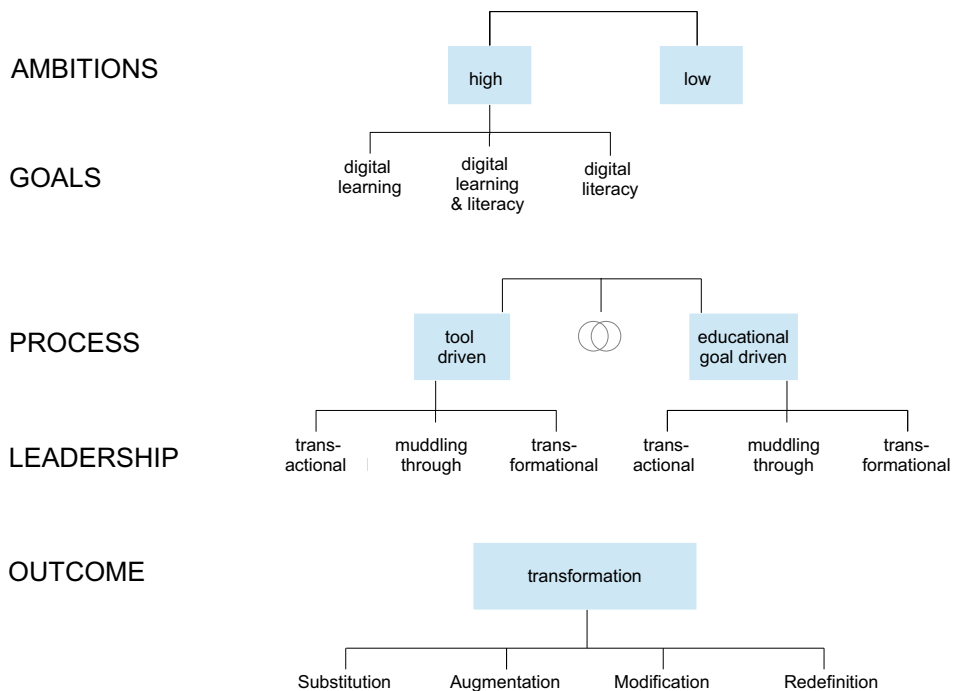


Figure 1. Level of digital awareness in upper secondary schools.

To ensure the reliability of the codes and categories, all interview transcripts were analyzed at least twice (both before and after the development of the coding guideline) and compared with one another. A second coder ensured intercoder reliability with a Cohen's kappa of 0.72. The categories with keywords and relevant statements made by principals helped to create an initial picture of principals' views and ideas. After a first tabulated analysis, we grouped the respondents into types of similar response patterns to be analyzed in more detail. Considering the literature on transformational and transactional leadership (Leithwood, 1992; Leithwood & Jantzi, 2000; Sun & Leithwood, 2012), the following criteria have been set to classify principals' leadership as transactional or transformational. We have categorized principals' statements in the interviews as rather transactional, if at least two of the following indicators were observed while indicators of transformational leadership (listed below) were missing: Principals literally mention 'general goals', 'incentives', 'rewards' or mere 'motivation' as a substantial element of their leadership and 'further training' as something everybody should attend. These categories were indicators for a rather directive and non-participatory style of transactional leadership. Principals' leadership was classified as rather transformational if they explicitly talked about at least one of the following aspects: 'intrinsic goals' or 'intrinsic motivation' for change, described themselves as models of change or emphasized their deep personal conviction and stake, or even passion, regarding digital change in their schools, or if they stressed the importance of an involvement of fellow teachers in the process. These categories were indicators for a rather open and participatory style of transformational leadership. If none of the above-mentioned criteria were met and principals' statements showed no clear goals or no clear leadership practices regarding digital change, their leadership was classified as 'muddling-through'.

Results and findings

All principals have clear perceptions about educational opportunities in the area of digitalization, even if many of their ideas are not implemented at their school (and will not be in the near future). Remarkably, all of the principals see digital technology as an opportunity in relation to individualized teaching and autonomous learning, irrespective of their leadership style. They believe that technical change or the integration of digital media will lead to student-oriented teaching. Several principals expressed the feeling that through digitalization, the role of teachers both will and must change. As two principals explained:

If you really want to realize this [digital development], there will be major changes in the way you teach. I say, once again, this will lead us into independent work on the part of the learner (Excerpt from Interview 7).

The way you teach will be different. And it's a challenge for teachers. They have to foster new [student] competencies. We have to adapt. So, there will be changes, there is no other way (Excerpt from Interview 1).

Table 1. Tabulated responses of school principals from the nine selected schools on leadership processes for technology integration.

	Ambitions	Goals	Process	Leadership	Outcome
School 1	High	Digital Literacy	Tool driven	Transactional	Substitution
School 2	High	Digital Literacy	Tool driven	Transactional	Substitution
School 3	High	Digital Literacy	Tool driven	Transactional	Substitution
School 4	High	Digital learning & literacy	Tool and educational goal driven	Transactional and transformational	Substitution/Augmentation
School 5	Low	Unclear	Unspecific	Muddling through	No outcome
School 6	High	Digital Literacy	Tool driven	Transactional	Substitution
School 7	High	Digital Literacy	Tool driven	Transactional	Substitution
School 8	High	Digital Literacy	Tool driven	Transactional	Substitution
School 9	Low	Unclear	Unspecific	Muddling through	No outcome

Changes are inevitable in principals' eyes. One principal remarked that 'if we don't do anything', schools could lose touch with the (digitalized) world. This would be unacceptable for students, 'because they have these devices' and they want to use them (Interview 9). Several principals expressed concrete ideas about what could change: Apps, tools or specific language software will redefine foreign language classes, to name one example. 'And actually, you need me as a foreign language teacher, what should I say, teachers could manage everything in a way that students could work by themselves' (Interview 6). Robots teaching languages would assist and eventually maybe replace teachers. However, these considerations remain quite theoretical. In practice, school principals greatly differ with regard to their actual ambitions in terms of planned developments in their schools. Table 1 provides an initial overview of the responses from the school principals in our case studies.

Principals see it as their obligation to prepare students for new challenges in a digital world, in one way or another. In the interviews, they used expressions such as 'duty', 'mandate' (Interview 1), 'responsibility' (Interview 4), or 'to empower students' (Interviews 3 and 5). In a digital world, students should be able to 'cope with life' (Interview 7). Nevertheless, there are substantial differences between the nine schools concerning principals' ideas and the integration of digital media. Leadership plays a decisive role.

Combining the criteria of our analysis, three digital leadership styles were identified (Table 2), where schools 5 and 9 represent style A (low ambitions, muddling through), schools 1, 2, 3, 6, 7, 8 are model examples of style B (high ambitions, transactional leadership) and school 4 represents the only example of style C (high ambitions, transactional and transformational leadership). These leadership styles will be discussed in detail.

Table 2. Digital leadership styles.

	Style A	Style B	Style C
Leadership	Muddling through	Transactional	Transactional and transformational
Ambitions	Low	High	High
Goals	Unspecific	Digital literacy	Digital learning and literacy
Process	Unclear	Tool driven	Tool and educational goal driven
Outcome	No outcome	Substitution	Substitution/Augmentation

Style A: muddling through, low ambition

Two schools belong to this group. Regarding digital transformation, the two principals do not say that they are not ambitious but they do not have concrete goals. Furthermore, neither school seems to meet the technical requirements to work with digital technologies on a large scale, as both principals said:

We have wireless internet in the whole building, but it isn't efficient' (Excerpt from Interview 5).

Our wireless internet is weak. We need infrastructural modifications (Excerpt from Interview 9).

A closer look at the interviews reveals that the reasons for this situation are quite different at these two schools. At the first school, teachers question digitalization. Teachers, like the principal, decided, that they do not want to become a 'school without paper' (Interview 5). Several teachers fear electromagnetic radiation. The principal mentioned that they check if it is possible to switch off wireless Internet in classrooms when it is not needed. Moreover, the principal emphasizes several times that teachers should not be forced to do anything specific just because of digital development. They should have a free choice, that is, voluntariness is important to this principal. In relation to administrative processes, however, teachers are supposed to use digital devices to enter grades, for example. The next steps are not very clear. The principal, undecided, mentions many ideas, some of which remain vague. The following quotation illustrates this point:

And then 'bring your own device'. What is this device? If it is a Smartphone, well, everybody has one, right? [...] I'd rather think of, more, really, a device with a keyboard, a personal working device for students. But I am skeptical as to whether we should really start next summer (Excerpt from Interview 5).

The principal of the other school remarked: 'We should deal with this topic in a more intense way. Teenagers have different views than we do' (Interview 9). However, there are no concrete plans. In the interview, the principal explains that most teachers in this school work part-time, which can be an obstacle. In other words: Either they do not have time for further education with regard to digital transformation or they see it as too time-consuming. The principal believes that if teachers identified with a specific goal and if they saw a benefit for students, they would be in. 'But time commitment is a problem' (Interview 9).

In summary, no specific digital development can be seen in this group. There is no outcome in view of the SAMR model. In terms of the individual context, the absence of clear goals and strategies formulated by principals can be explained in different ways. Instead of concrete objectives, the two principals mentioned 'problems' and spoke in the interviews about hesitant teachers. One also mentioned infrastructural deficits and personal doubts, while the other referred to staff insecurities that might hinder digital transformation. Both principals' leadership around digital change can be regarded as undecided or unclear. One principal (Interview 9) is explicitly dissatisfied with this situation.

Style B: transactional leadership, high ambition

The majority of schools in this study, that is, six, belong to this group. All six principals stress that digital development is important to them and that their school is ambitious. Their leadership can be characterized as transactional: It is about agreements and rewards.

Principals indicate that they encourage committed teachers and give them autonomy to realize their ideas. As one principal emphasized: 'It's okay if teachers have ideas and want to try things out' (Interview 1). On the other hand, there are teachers unwilling to work with digital technology who can hinder digital transformation in school. All principals state that they talk to non-committed teachers and give them time. They try to 'get them on board' (Interview 8) but also warn them that they might lose connection or even lose their jobs. One principal, however, admits that he does not want to discipline an employee: 'Teachers have to be authentic; I am not willing to correct their behavior if they can explain to me why their teaching method is suitable for their subject' (Interview 2). All principals in this group are convinced that, sooner or later, teachers have to adapt to this digital development because they cannot prevent it. Furthermore, they agree that young teachers are important when it comes to supporting digital change. In relation to digital change, goals are essential for all principals. As two principals mentioned:

I've formulated a goal for the team. The goal of this year is not to just use digital technologies for data storage, but actually work with them. And to integrate digital technologies in our classes. That was my welcome letter and I've sent it to every teacher (Excerpt from Interview 2).

It is important that the school has a goal. And one goal is that all students have a device in class (Excerpt from Interview 7).

All six principals indicate that they want to involve teachers in the process of digital change. They mention 'steps that have to be taken'. The 'first step' can be a 'further training session' (Interview 3). Four principals emphasize that they are going to organize further training sessions for teachers and they explain that they will do it soon (Interviews 1, 2, 3, 7). All of them state that they are personally interested in digital technologies. One principal would go further and mention that technology can be personally inspiring: 'I am a PC freak' (Interview 2). Another principal, however, says that he is not technologically gifted. All principals agree that they feel some sort of obligation: Some of them assume that with respect to digital development, 'Switzerland has fallen behind and other countries are one step ahead' (Interview 3). Others believe that there is 'pressure' because other schools in the area are ahead (Interview 8). All principals are convinced that upper secondary schools need to prepare students for the job market or university.

Upper secondary schools [...] can't say: 'We aren't part of it'. Impossible. It's our job to prepare students for the next, the next level of [tertiary] education. And this means that they have to be educated in how to deal with digital media (Excerpt from Interview 1).

This quotation describes principals' goals and concerns digital literacy. Five principals emphasize that one important goal is for students to learn how to 'deal with' digital technologies. Students, as well as teachers, should be able to 'handle' digital devices

(Interviews 6 and 7) and become ‘media-savvy’ (Interviews 3 and 8) or competent with technology. One principal explains that prevention of online risks is important: ‘There is bullying, cyberbullying, it’s a big issue’ (Interview 2). Furthermore, all principals talk about independent learning in the future. Three principals specifically mention experiences with ‘e-learning’ in connection with ‘tools’ (Interviews 2, 7 and 8).

The process of digital development is mainly tool driven and it is important to all principals that their school has the right infrastructure. They see a good ‘server’ infrastructure with a stable connection as a basic requirement (Interviews 6 and 7). Generally, tools are at the center: Software, a ‘school software’ or ‘administration software’ (Interview 1), numerous apps or digital platforms for teachers and students, such as ‘One Note’ (Interviews 1 and 6) or ‘Dropbox’ (Interview 8), as well as a projector, a ‘beamer’ and a ‘presenter’ (Interview 1) or a digital whiteboard or a ‘smart board’ (Interviews 2 and 8) in the classrooms are mentioned. All principals talk about BYOD: ‘Bring your own device’ has been or is going to be introduced at their schools. Here are three examples:

Two years ago, we introduced digitalization at school. All classes have laptops or ‘bring your own device’. Students bring their device to school. Teachers do not give handouts. Students can complete every task on their device (Excerpt from Interview 2).

So, we started with ‘bring your own device’. We evaluated it, we gained insights and we upgraded our internet. There is high-performance wireless internet in the whole school building now (Excerpt from interview 1).

We want to start earlier. Grade 10. ‘Bring your own device’ will be compulsory. Now we have to be prepared, technically and pedagogically (Excerpt from Interview 3).

Even though their general style of leadership may be considered transactional, principals care for their teachers. They talk about teachers’ ‘fears of contact [with digital technology]’ (Interview 2) and that teachers’ ‘concerns have to be taken seriously’ (Interview 6). As one principal stated: ‘We have to accompany and support teachers in this process and we also have to be aware of their fears and insecurities and to listen to them’ (Interview 8).

With regard to the SAMR model, the outcome can mostly be described as ‘Substitution’. All principals stress that digital technology should serve as a tool, for instance, to save teachers’ handwritten notes digitally. Two principals mention the idea of students doing essays or exams on their computer (Interviews 1 and 7):

And the test, it is an essay. The aim is that you can write it on your notebook. And after a little while, we’ll have the situation that you more or less write the whole exam on your notebook. But that will need some preparation (Excerpt from Interview 1).

Teachers, on the other hand, have to use a specific tool to save students’ grades and note when students are absent. All principals talk about the co-existence of digital and analog media. Two principals, however, also discuss ideas that can be described as an improvement for class (‘Augmentation’): One principal mentioned that in some cases, digital technologies can be a real help, for instance, when students do online research in history. The other explains that students benefit from online discussion forums: ‘But they actually don’t use it. You have to force them to do so. And then they do – against their will’ (Interview 2).

Style C: transactional and transformational leadership, high ambition

Only one school belongs to this group. The principal's leadership can be regarded as transactional and transformational at the same time. Target agreements or goals are essential. Beyond that, it is important to this principal that teachers, as well as students, 'do it on their own', just 'out of interest' (Interview 4). This principal talks about 'many ideas' regarding digital development but doesn't want to 'do it top down' because 'it has to be bottom up':

So, no BYOD-project! I don't want to scare people (Excerpt from Interview 4).

Instead, this principal has another 'strategy', focusing on a few committed teachers: 'And so I support these teachers'. (Interview 4). The principal calls 'these' digital-savvy teachers 'pilots', who should 'not be stopped' because they motivate others:

When I was young, I had supervisors who just let me do my job. They didn't hold me back. So, now, I just have to let it go on working. I do not have to interfere, as long as it fits our strategy. A lot of things are happening. Distrustful teachers! And hey, now I see that they are interested. Last week I noticed this one teacher complaining about digital media. Immediately, his colleague opened his computer and showed how easy it is. Now this teacher wants to try this software out on his own (Excerpt from Interview 4).

'Pilots' are important to this principal. One of them is going to organize internal further training courses. It is important for this principal to get teachers involved in this process, in different ways. In the meantime, he wants to be part of it:

We have a biology teacher and a math teacher and me as a principal who finished a Certificate of Advanced Studies [on educational technology] last Friday. Now we're doing this individual work and we'll present these lessons to our team soon (Excerpt from Interview 4).

In terms of digital transformation, goals include digital learning and literacy. The process can be described as driven by tools and educational goals. This principal wants students to 'get an introduction into the subject of digitalization' (Interview 4). The goal is that students develop their mind-set and think about data volumes, algorithms and how to write a software program. They should ask questions about the performance of powerful software products and the consequences for their personal lives. Another goal is that students and teachers alike, build digital networks and work together in small interdisciplinary groups in which they should develop their projects.

The principal emphasizes that students must 'have tools that fit, where they make progress regarding their own learning and working efficiency' (Interview 4). There is no concrete 'bring your own device' policy. However, students and teachers all bring their devices to class. Teachers have to use Microsoft Office tools such as Teams or Sway to digitally communicate with the principal and with each other.

When dealing with resistance, this principal describes his approach as follows: 'We have teachers who teach in a traditional way and I won't force them [to work with digital media]' (Interview 4). Still, he reports that all teachers are somehow interested in digital media and all of them want to be involved in the process of digital development in one way or another.

In relation to the SAMR Model, the outcome can be described as 'Substitution' and 'Augmentation': Teachers use tools to make documents available to students. Beyond that, digital media support and foster self-organized learning. This includes individual learning as well as learning collaborations.

Discussion and conclusion

The aim of this study was to explore upper secondary school principals' perspectives toward digital change and to show how they act and react with regard to the transformation process. The analysis was undertaken to provide insights into the interplay between principals' views on technology integration goals, self-described leadership practices and digital change in schools, reported by principals. It also provides insights in digital changes expected to happen in the future. Supporting previous studies, the results demonstrate that principals are indeed key players for digital development in schools (Davies, 2010; Dexter et al., 2016). However, the quality of digital change is not only dependent on goals or goal clarity, but also on the style of leadership (Dexter & Richardson, 2020; Leithwood & Jantzi, 2000). One of the interviewed principals applies a rather transformational as well as a transactional leadership style: This principal reported more rapid and more profound changes than all the other principals (oriented toward a more transactional type of leadership). This principals' leadership style can be associated with faster implementation of digital technologies. Furthermore, the focus in this case was educational goal driven rather than tool driven (see, Figure 1). Research shows that transformational leadership requires a certain amount of leeway (Bolívar & Moreno, 2006). When pressured with the introduction of a new digital curriculum, school principals might resort to more traditional transactional leadership activities. Six principals' leadership styles can be characterized as transactional. They mainly focus on the preparations for the implementation of digital literacy in the curriculum. This was a mandatory project for all schools of this region in the timeframe of our study. The process at these six schools is tool driven, and they concentrate on technology, rather than on educational goals. Thus, digital technology mainly serves as an add-on with little overarching effect (Cuban, 2001; McLeod, 2018; Puentedura, 2015). Two principals have unspecific goals. Their leadership style can be described as 'muddling through' or 'laissez-faire' despite the necessities of the new curriculum. The process at their school is unclear. With regard to the SAMR model, no technology-supported transformation can be seen in relation to teaching and learning. Apart from these two principals, all principals interviewed state that they have high-level and long-term goals in the area of digital development. As digital leaders, they must have a well-defined and transparent strategy as well as a clear picture of the characteristics and the needs of their educational institution (Cortellazzo et al., 2019). Regardless of their ambitions, all nine principals interviewed acknowledge the importance of teachers identifying with the goals (Judge & Piccolo, 2004). This means that principals' actions are well thought out. It also means that major changes do not happen from one day to the next. High-level goals correspond positively with digital change. With regard to the SAMR model, however, the extent of digital transformation of teaching and learning doesn't exceed the first level and technology basically serves as a direct tool substitute without functional change (Puentedura, 2012). An exception is the school with a principal oriented toward a transactional *and* transformational leadership style. In this school, the integration of digital technology goes beyond substitution. Teachers use digital media to improve students' educational experiences.

Summing up these results, our study seems to support the observation in previous case studies that transformational leadership practices are used to accomplish more profound change (Afshari et al., 2009; Hadjithoma-Garstka, 2011; Hauge & Norenes, 2015;

Lindqvist, 2019; Ng, 2008; Schrum & Levin, 2013; Yuen et al., 2003). Transactional leadership practices seem to be preferred by school leaders when the envisioned change is more superficial. Although we were not able to study the actual impact of these leadership activities on teacher practices and student learning, we were able to show in the interviews with school principals that more ambitious goals are enacted with more comprehensive and participatory leadership activities.

Limitations

Switzerland is known to be quite conservative when it comes to technology integration in schools, which can be regarded as a limitation. For historical reasons, Swiss upper secondary schools have a lot of autonomy. In Switzerland's highly stratified education system, where the majority of students do an apprenticeship after secondary school, upper secondary schools are (still) seen as elite schools. Their leaving certificate qualifies young people for university entrance, and students have open access to all universities in Switzerland, with no entrance test except for medical degrees. In this context, Swiss upper secondary school teachers are quite autonomous when it comes to teaching methods. All nine principals interviewed convey the impression that they are well aware of that, regardless of their digital leadership style as discussed in the last section of this paper. All of them stress that they respect teachers' autonomy.

The population in this study can be seen as another limitation. Only one of the nine principals interviewed described substantial digital transformation processes in school. The connection to a transformational leadership approach has been discussed in this paper. However, this principal might not be representative of the greater population of upper secondary school principals in Switzerland. A future study might explicitly focus on transformational leadership and consider principals who would describe themselves as transformational leaders. This would make it possible to get a clear and differentiated picture of the influence of transformational leadership on digital change at school.

Lastly, it should be pointed out that the data was collected in 2019 and so it predates Covid-19. The principals, not at that point experiencing any pressure, talked openly about their views of digital development in education. It cannot be ruled out that they might modify their statements if interviewed again. It has been assumed that the rapid changes during the pandemic have changed leadership practices (Harris & Jones, 2020). However, best practices of leadership during crisis seem to be closely aligned to approaches of distributed and transformational leadership (Fernandez & Shaw, 2020). Future studies might therefore specifically focus on principals reporting changes in their leadership behavior due to the Covid-19 situation.

Implications

This study makes an important contribution to understanding digital development in upper secondary schools in Switzerland. The discussion of the three digital leadership styles shows how principals cope with chances and challenges and how their actions promote (or hinder) digital change.

The presented typology ('level of digital awareness') can be understood as an overview of concepts and attitudes, but also of principals' ideas and theories in the area of digital development at upper secondary schools. It demonstrates how digital leadership can be characterized. The classification to the three groups and the three leadership styles discussed is clear. The six schools or principals in group B (transactional leadership) show similarities in terms of their ambitions, goals, process and outcome. In contrast to the homogeneity in group B, the two schools in group A (leadership as muddling through) are harder to compare due to different contexts. Only one school belongs to group C (transactional and transformational leadership). Nevertheless, there are strong disparities between these groups. The shape of digital leadership in the three groups is fundamentally different.

The implication for policy-makers is that in the area of digitalization, principals' views are crucial. Their leadership style affects digital transformation in their school, and it has an influence on goals, process and outcome. Transformational leadership is more effective and means a faster integration than Transactional Leadership. A top-down approach may not work well or has only limited impact at school level, since principals' digital leadership is participatory. Reforms are only successful when supported by principals.

Another implication is that in principals' eyes, technical change or the integration of digital media should clearly lead to student-oriented teaching. Educational goals are paramount to raising public support in the area of digital development in school. However, the study also raises expectations in relation to educational opportunities with digital technologies. In order to succeed with digital transformation, teachers must be prepared for work in a digital environment. Teacher education needs to address this responsibility and further research is required to explore ways to undertake this task.

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Notes on contributors

Michael Ruloff is Professor at School of Education, University of Applied Sciences and Arts Northwestern Switzerland. His research focuses on teachers' and student teachers' professional development. In particular, his research examines the way digital technology is implemented in schools and universities.

Dominik Petko is Professor at the Institute of Education, University of Zurich, Switzerland. His research areas include educational technology as well as technology related skills and beliefs of teachers. A particular focus of the research is on the use of digital technology in teaching and on effective approaches to teacher education, especially at upper secondary schools.

ORCID

Michael Ruloff  <http://orcid.org/0000-0002-6947-3493>

Dominik Petko  <http://orcid.org/0000-0003-1569-1302>

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